AWS Monitoring and Alerting Assessment

About lab assessment

Lab assessments are Hands-on labs with the gloves off. You jump into an auto-provisioned cloud environment and are given a goal to accomplish. No instructions, no hints. To pass, you'll have a limited time to demonstrate your problem-solving skills and get the checks that inspect the state of your lab environment.

Description

Amazon Web Services gives you the tools you need to identify, troubleshoot, and alert you on issues when they occur. By learning how to use monitoring and alerting services effectively, you will be more productive in diagnosing and fixing operational problems.

This hands-on lab challenge will test your practical ability to modify a solution in a production-like AWS environment. You will be presented with a task and set of requirements you must fulfill to pass the challenge.

This is a real environment, which means you can prove your knowledge in an applied way, substituting multiple choice questions for a dynamic performance-based exam situation.

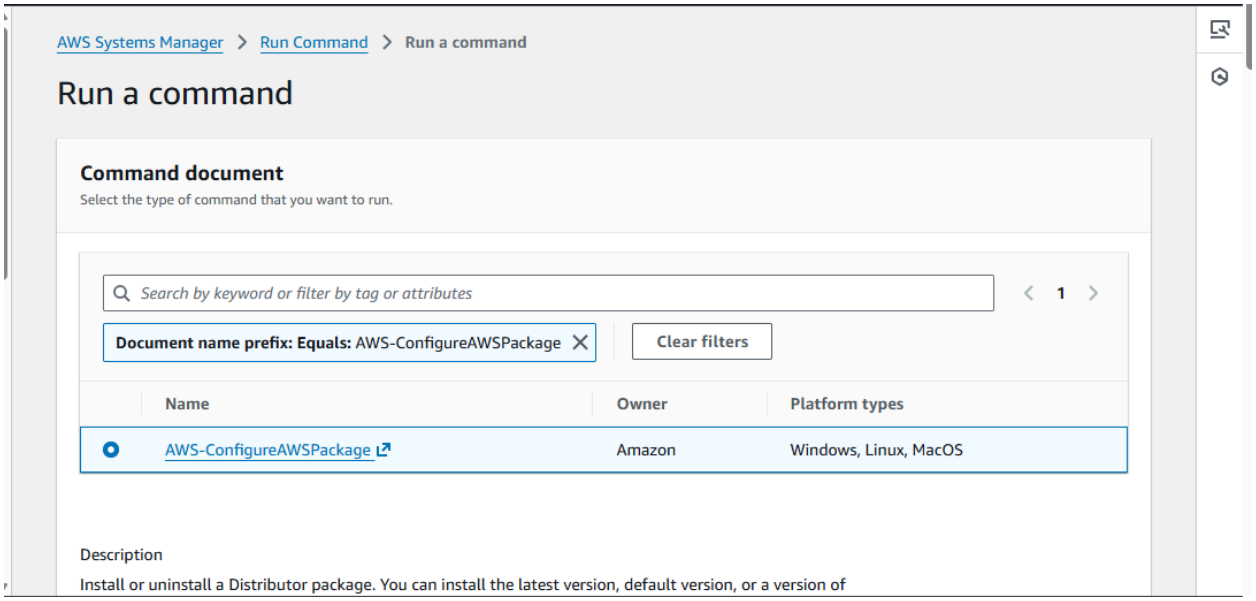
Prerequisites

- Using SQS, SNS and SES in a Decoupled and Distributed Environment (Course)
- Managing Your Instances using the AWS Systems Manager Run Command, Documents, & Parameter Store (Course)
- How to Implement & Enable Logging Across AWS Services (Part 1 of 2) (Course)
- Compute Fundamentals For AWS (Course)

Level 2: Install the CloudWatch Agent Using SSM

Run an SSM command that satisfies the following:

- Uses the **AWS-ConfigureAWSPackage** document
- Installs the **AmazonCloudWatchAgent** package
- Uses the **latest** version of the package
- Targets EC2 instances with the tag:
  - Key: **Name**
  - Value: **cloudacademylabs**
- Does not write output to an S3 bucket



Command parameters

Action
(Required) Specify whether or not to install or uninstall the package.

Install

Installation Type
(Optional) Specify the type of installation. Uninstall and reinstall: The application is taken offline until the reinstallation process completes. In-place update: The application is available while new or updated files are added to the installation.

Uninstall and reinstall

Name
(Required) The package to install/uninstall.

AmazonCloudWatchAgent

Version
(Optional) The version of the package to install or uninstall. If you don't specify a version, the system installs the latest published version by default. The system will only attempt to uninstall the version that is currently installed. If no version of the package is installed, the system returns an error.

latest

Additional Arguments
(Optional) The additional parameters to provide to your install, uninstall, or update scripts.

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Simple Notification Service | EC2 | Aurora and RDS | Lambda | S3 | Athena | Step Functions | DynamoDB | Simple Queue Service | Amazon OpenSearch Ser... | Kinesis | VPC

AWS Systems Manager

- Review node insights
- Explore nodes
- Diagnose and remediate
- Just-in-time node access New
- Settings

Node Tools

- Compliance
- Distributor
- Fleet Manager
- Hybrid Activations
- Inventory
- Patch Manager

Target selection

Target selection
Choose a method for selecting targets.

- Specify instance tags — Specify one or more tag key-value pairs to select instances that share those tags.
- Choose instances manually — Manually select the instances you want to register as targets.
- Choose a resource group — Choose a resource group that includes the resources you want to target.

Specify instance tags
Specify one or more instance tag key-value pairs to identify the instances where the tasks will run

Tag key	Tag value (optional)	
Name	cloudacademylabs	Add

Enter a tag key and optional value applied to the instances you want to target, and then choose **Add**.

Other parameters

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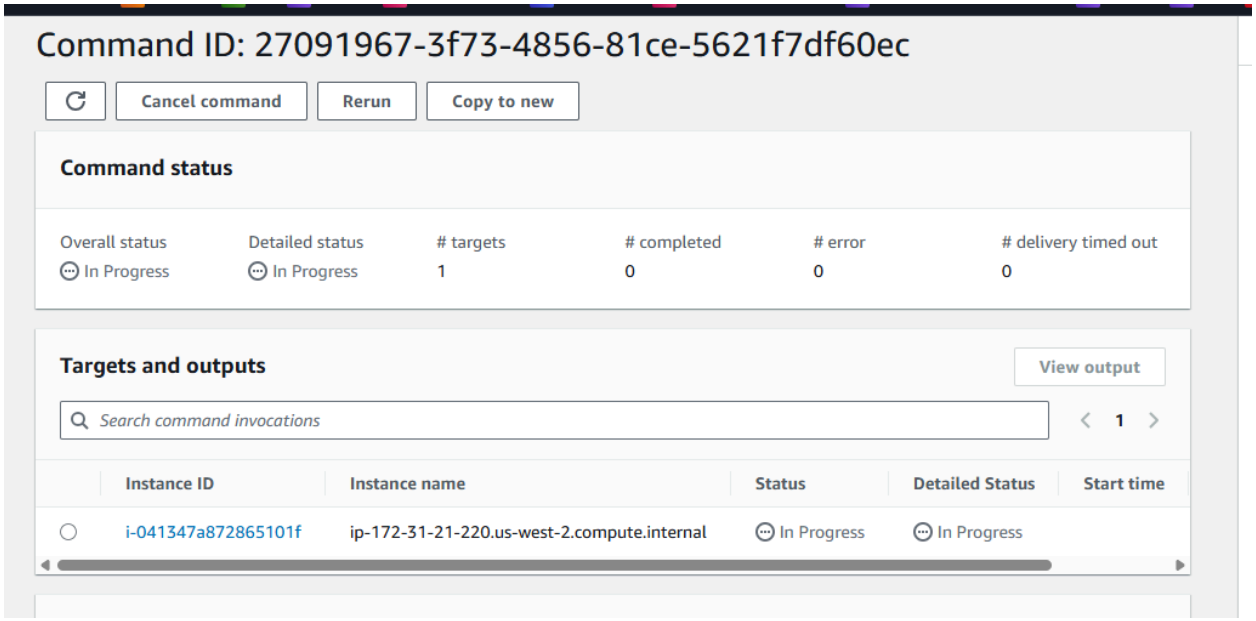
Output options

Write command output to an Amazon S3 bucket
Write all command output to an Amazon S3 bucket. Command output in the console is truncated after 24,000 characters.

☐ Enable an S3 bucket

Send command output to Amazon CloudWatch logs
You can stream and encrypt log data for all commands in your account to a CloudWatch Logs log group in your account. Learn more

☐ Enable CloudWatch logs



Level 2: Configure the CloudWatch Agent

Run an SSM command that satisfies the following:

- Uses the **AmazonCloudWatch-ManageAgent** document
- Uses the **configure** action in **EC2** mode
- Uses **ssm** for the configuration source
- Uses **/lab/cloudwatch/configuration** for the configuration location
- Targets EC2 instances with the tag:
  - Key: **Name**
  - Value: **cloudacademylabs**
- Restarts the agent after configuration
- Does not write output to an S3 bucket

Action
The action CloudWatch Agent should take.

configure

Mode
Controls platform-specific default behavior such as whether to include EC2 Metadata in metrics.

ec2

Optional Configuration Source
Only for 'configure' related actions. Use 'ssm' to apply a ssm parameter as config. Use 'default' to apply default config for amazon-cloudwatch-agent. Use 'all' with 'configure (remove)' to clean all configs for amazon-cloudwatch-agent.

ssm

Optional Configuration Location
For configure actions when optionalConfigurationSource is ssm: Single SSM parameter name, or comma-separated list of SSM parameter names to merge multiple configurations.

/lab/cloudwatch/configuration

Optional Restart
Only for 'configure' related actions. If 'yes', restarts the agent to use the new configuration. Otherwise the new config will only apply on the next agent restart.

yes

Target selection

Target selection
Choose a method for selecting targets.

- (selected) **Specify instance tags** — Specify one or more tag key-value pairs to select instances that share those tags.
- Choose instances manually — Manually select the instances you want to register as targets.
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Specify instance tags
Specify one or more instance tag key-value pairs to identify the instances where the tasks will run

Tag key | Tag value (optional) | Add

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Name : cloudacademylabs ✕

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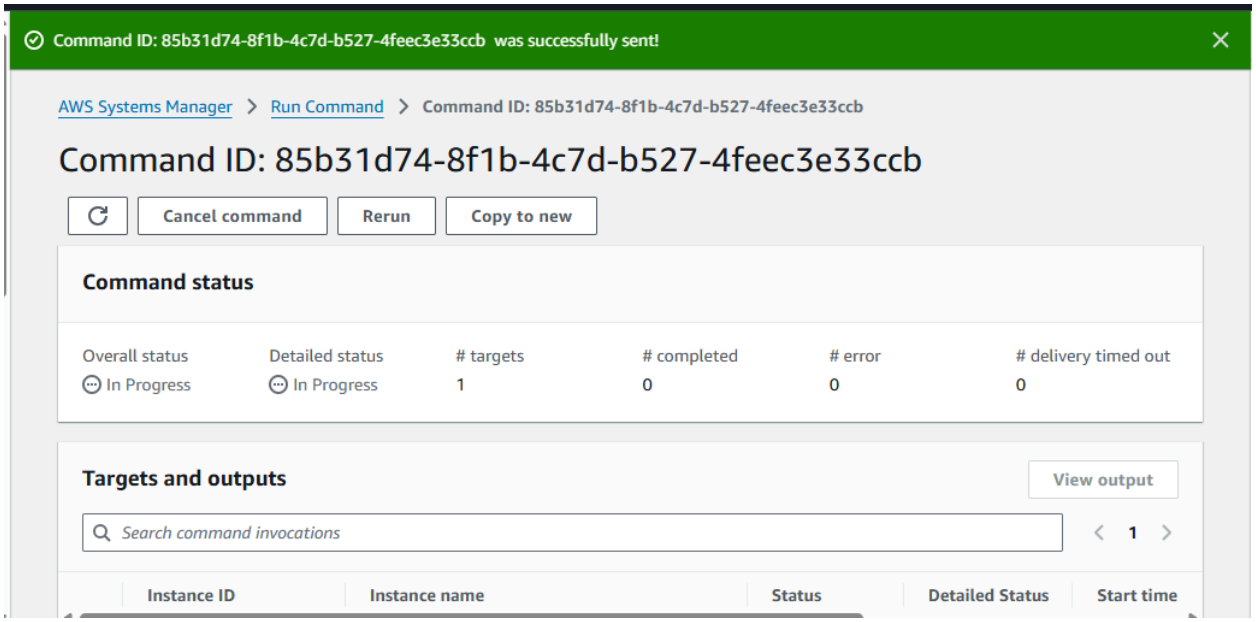


Level 3: Create a CloudWatch Alarm

Create an alarm in Amazon CloudWatch that satisfies the following:

- Uses the metric that:
  - Is in the **CWAgent** namespace
  - Is named **mem_used_percent**
  - On a running instance with the name tag **cloudacademylabs**
- Uses the **Maximum** statistic over a **1 minute** period
- Is triggered when metric is over 50 percent
- Treats missing data as missing
- Sends a notification to your SNS topic

Note: The agent's data won't appear immediately. If you don't see a metric for a running instance, wait one or two minutes and retry.

CloudWatch > Alarms > Create alarm

Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Metric

Edit

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.

Namespace
CWAgent

Metric name
mem_used_percent

InstanceId
i-041347a872865101f

ImageId
ami-02373b999c076305d

InstanceType
t2.micro

Statistic
Maximum

Period
1 minute

Conditions

Threshold type

- (●) **Static** — Use a value as a threshold
- ( ) **Anomaly detection** — Use a band as a threshold

Whenever mem_used_percent is...
Define the alarm condition.

- ( ) Greater — > threshold
- (●) Greater/Equal — >= threshold
- ( ) Lower/Equal — <= threshold
- ( ) Lower — < threshold

CloudWatch > Alarms > Create alarm

than...
Define the threshold value.

50

Must be a number.

▼ **Additional configuration**

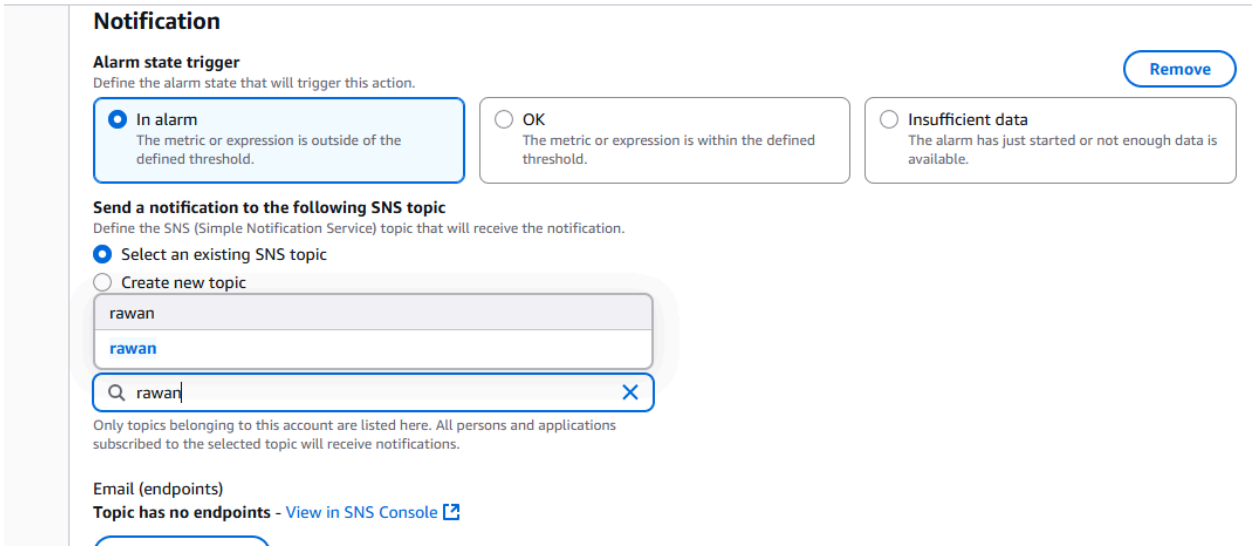
Datapoints to alarm
Define the number of datapoints within the evaluation period that must be breaching to cause the alarm to go to ALARM state.

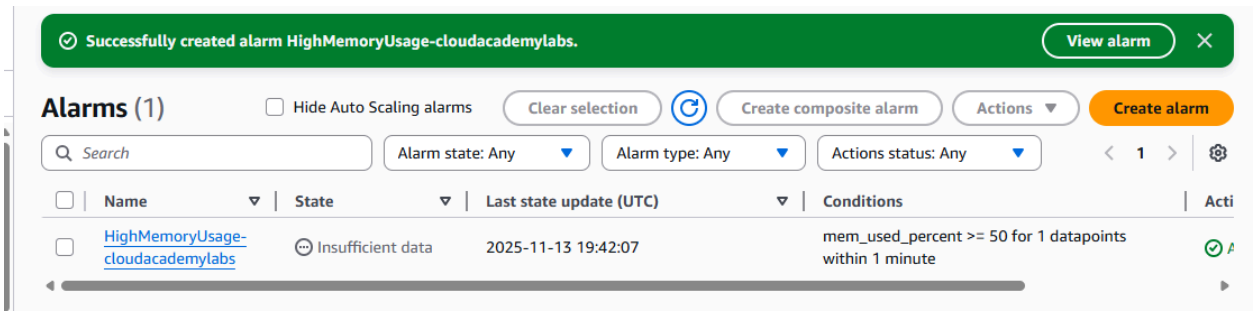
1 out of 1

Missing data treatment
How to treat missing data when evaluating the alarm.

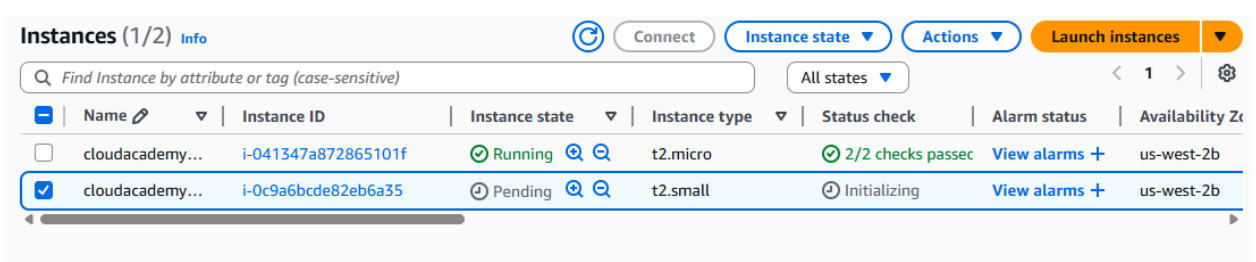
Treat missing data as missing

Cancel Next





Level 3: Replace the EC2 Instance

Using the existing launch template, launch a new EC2 instance with the **t2.small** instance type and terminate the existing **t2.micro** instance

Note: This check will not pass until the new instance is in the **Running** state and the old instance is **Terminated**